

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

Year 11

2 Unit

July 2010

Common Test

Time allowed: 70 mins

Name: _____

Teacher: _____

Instructions:

- Begin each question on a new page
- Marks shown are approximate and may be varied
- Show necessary working
- Full marks may not be awarded if working is poorly set out or difficult to read

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
7	/7	/8	/7	/8	/8	/7	/8	/60

QUESTION 1**MARKS**

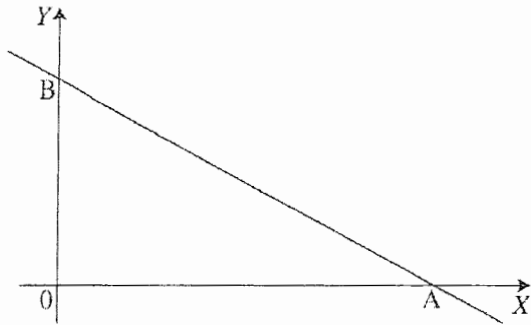
- (a) Solve for x : $\frac{3}{x-2} = \frac{4}{2x+5}$. 1
- (b) Express $0.6\overline{81}$ as a fraction in simplest terms. 2
- (c) Use your calculator to find the value of $\frac{(1.49)^2 - 1.98}{\sqrt{11.62 + 8.34 \times 2.72}}$ correct to 3 significant figures. 2
- (d) Simplify $\frac{2x^3 + 128}{x^2 - 9} \div \frac{x+4}{x+3}$. 2

QUESTION 2 (*Start a new page*)

- (a) Solve $|7 - 3x| = 4$. 2
- (b) The lines $ax + 2y = 6$ and $4y = bx - 9$ are parallel. Find the value of $\frac{a}{b}$. 3
- (c) Solve $6x^2 = x + 2$. 2

QUESTION 3 (Start a new page)

(a)

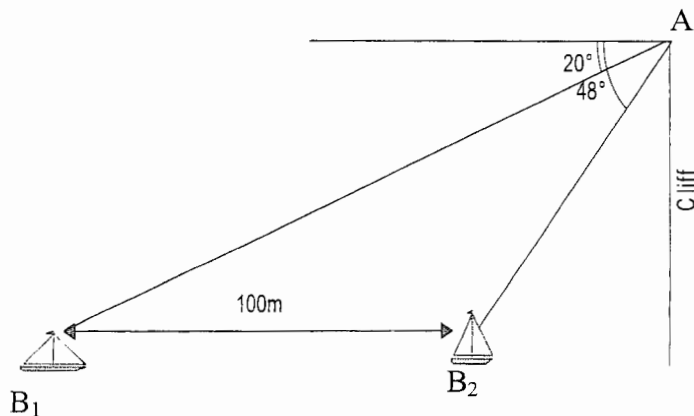


B is the point $(0, 6)$ and A is a point on the x -axis.

- (i) If $AB = 10$ units, find the coordinates of A . 1
 - (ii) Find the coordinates of M , the mid-point of AB . 1
 - (iii) Show that AB has equation $3x + 4y - 24 = 0$. 2
 - (iv) Find the shortest distance from O to the line AB . 2
- (b) Solve the simultaneous equations $2x - 3y - 12 = 0$ and $5x + 2y - 11 = 0$ 2

QUESTION 4 (Start a new page)

- (a) From the top of a cliff an observer measures the angles of depression of two boats on a lake, one behind the other in a straight line to be 20° and 48° . The boats are exactly 100 m apart.
- i. Find A to B_2 2
 - ii. Find the height of the top of the cliff above sea level (to the nearest metre). 2

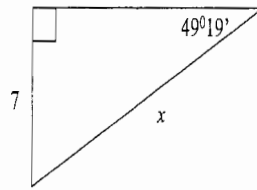


(b) Given $f(x) = 5 - x^2$, find $f(-2)$

1

(c) Evaluate x to one decimal place.

2



QUESTION 5 *(Start a new page)*

(a) State the domain and range of the function $y = -\sqrt{16 - x^2}$.

2

(b) Given $\sin \theta = \frac{\sqrt{3}}{10}$ and $\tan \theta < 0$, find the exact values of $\cos \theta$ and $\cot \theta$.

2

(c) Solve $\sin^2 \theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.

2

(d) $4^{x+2} \times 2^{2x-3} = 8^x$

2

QUESTION 6 (*Start a new page*)

(a) If $\sin \alpha = 0.6$ and $0^\circ < \alpha < 90^\circ$ write down the exact value of:-

i. $\tan (180^\circ - \alpha)$ **1**

(b) Find the exact value of $\sin 225^\circ$ **1**

(c) A function is defined by the rule:

$$f(x) = \begin{cases} x+2, & x \geq 0 \\ \frac{1}{x}, & x < 0 \end{cases}$$

Find:

(i) $f(3)$ (ii) $f(-2)$ **2**

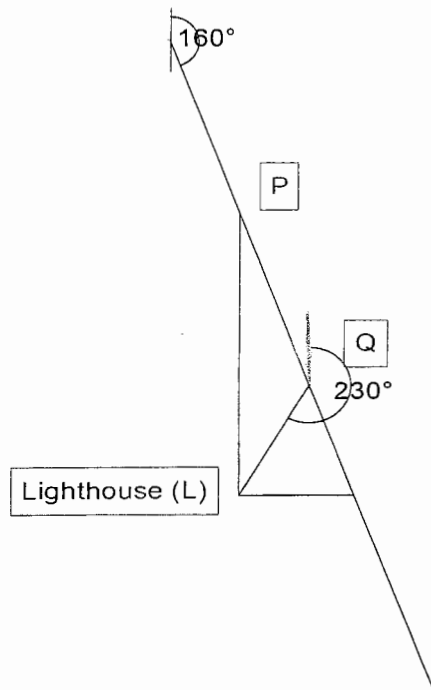
(d) Solve for $-180^\circ \leq \theta \leq 180^\circ$

i. $\sin \theta = -\frac{\sqrt{3}}{2}$ **2**

ii. $\tan \theta + 1 = 0$ **2**

QUESTION 7 (Start a new page)

A boat sailing on a bearing of 160° at 12 km/h arrives at a point P due north of a lighthouse L at precisely noon. Forty minutes later the boat is at a point Q where the lighthouse is on a bearing of 230° .



- | | |
|---|---|
| (i) Copy the diagram onto your book and find the distance PQ . | 1 |
| (ii) Calculate the size of angle PQL . | 1 |
| (iii) Hence calculate the distance PL to the nearest 100 m. | 2 |
| (iv) Find the time to the nearest minute, at which the lighthouse will be due west of the boat. | 3 |

QUESTION 8 *(Start a new page)*

(a) Prove the identity $(\sec \theta - \cos \theta)^2 = \tan^2 \theta - \sin^2 \theta$. **2**

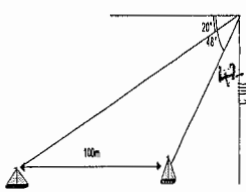
(b) If $x + y = 1$ show that $(x^2 - y^2)^2 + xy = x^3 + y^3$. **2**

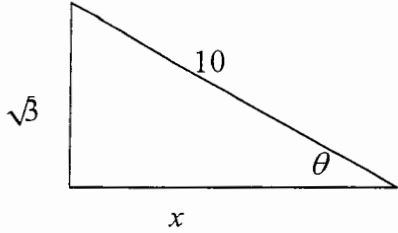
(c) (i) If $u = 3 + \sqrt{7}$ show that $u + \frac{2}{u} = 6$. **2**

(ii) Hence evaluate $u^2 + \frac{4}{u^2}$. **2**

CORRECTED SOLUTIONS

QUESTION 1 (a) $\frac{3}{x-2} = \frac{4}{2x+5}$ $3(2x+5) = 4(x-2)$ $6x + 15 = 4x - 8$ $2x = -23$ $x = \frac{-23}{2}$	QUESTION 2 (a) $7 - 3x = 4$ $7 - 3x = 4 \quad \text{or} \quad -7 + 3x = 4$ $3x = 3 \quad \quad \quad 3x = 11$ $x = 1 \quad \quad \quad x = \frac{11}{3}$
(b) Let $0.\overline{681} = x$ Then $10x = 6.818181818181\dots$ $1000x = 681.8181818181\dots$ $990x = 675$ $x = \frac{675}{990}$ $x = \frac{15}{22}$ (c) 0.040 993 $\therefore 0.0410$ 3 significant figures	(b) $ax + 2y = 6$ $2y = -ax + 6$ $y = -\frac{ax}{2} + 3 \quad \therefore \text{Gradient} = -\frac{a}{2}$ $4y = bx - 9$ $y = \frac{bx}{4} - \frac{9}{4} \quad \therefore \text{Gradient} = \frac{b}{4}$ Since lines are parallel, gradients are equal. $\therefore -\frac{a}{2} = \frac{b}{4}$ $\therefore \frac{a}{b} = \frac{-2}{4} = \frac{-1}{2}$
(d) $\frac{2x^3 + 128}{x^2 - 9} \div \frac{x + 4}{x + 3}$ $= \frac{2(x^3 + 64)}{x^2 - 9} \times \frac{x + 3}{x + 4}$ $= \frac{2(x + 4)(x^2 - 4x + 16)(x + 3)}{(x - 3)(x + 3)(x + 4)}$ $= \frac{2(x^2 - 4x + 16)}{x - 3}$	(c) $6x^2 = x + 2$ $6x^2 - x - 2 = 0$ $(3x - 2)(2x + 1) = 0$ $3x - 2 = 0 \quad \text{or} \quad 2x + 1 = 0$ $x = \frac{2}{3} \quad \text{or} \quad x = -\frac{1}{2}$

QUESTION 3 (i) $AB^2 = OA^2 + OB^2 \text{ (Pythagoras' Theorem)}$ $100^2 = OA^2 + 36$ $OA^2 = 64$ $OA = 8$ $\therefore$ Coordinates of A are (8, 0)	QUESTION 4 (a) 
(ii) $x_M = \frac{8+0}{2} \qquad y_M = \frac{0+6}{2}$ $\therefore M(4, 3)$	i $x \div \sin 20^\circ = 100 \div \sin 28^\circ$ $x = 100 \times \sin 20^\circ \div \sin 28^\circ$
(iii) Equation of AB: $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ $\frac{y - 0}{x - 8} = \frac{6 - 0}{0 - 8}$ $\therefore y = -\frac{6}{8}(x - 8)$ $8y = -6x + 48$ $3x + 4y - 24 = 0$	$x = 72.85 \dots$ ii $\cos 42 = \text{cliff} \div 72.85$ $\text{cliff} = \cos 42 \times 72.85 \dots$ cliff = 54.138 m
(iv) $d = \left \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right$ $= \left \frac{-24}{\sqrt{9+16}} \right $ $= \frac{24}{5}u$	(b) $f(x) = 5 - x^2$ $f(-2) = 1$
$2x - 3y - 12 = 0 \dots (1) \dots \times 5$ $5x + 2y - 11 = 0 \dots (2) \dots \times 2$ $10x - 15y - 60 = 0 \dots (3)$ $-(10x + 4y - 22 = 0)$ $-19y - 38 = 0$ $y = -2 \text{ sub.} \dots (1)$ $\therefore x = 3$	(c) $\sin 49^\circ 19' = \frac{7}{x}$ $x = \frac{7}{\sin 49^\circ 19'}$ $x = 9.23 \dots$ $x = 9.2$

QUESTION 5 (a) Domain: $-4 \leq x \leq 4$ Range: $-4 \leq y \leq 0$	QUESTION 6 (a) $\alpha = -0.75$
(b)  $(\sqrt{3})^2 + x^2 = 10^2$ $x = \sqrt{97}$ and $\tan \theta < 0, \therefore 2nd \text{ quadrant}$ $\cos \theta = \frac{-\sqrt{97}}{10}$ $\cot \theta = \frac{-\sqrt{97}}{\sqrt{3}}$	(b) $\sin 225^\circ = \sin (180^\circ + 45^\circ)$ $= -\frac{1}{\sqrt{2}}$
(c) $\sin^2 \theta = \frac{1}{2}$ $\sin \theta = \frac{1}{\sqrt{2}}$ $\theta = 45^\circ \quad 135^\circ \quad 225^\circ \quad 315^\circ$	(c) $f(3) = 5$ $f(-2) = -\frac{1}{2}$
(d) $4^{x+2} \times 2^{2x-3} = 8^x$ $2^{2x+4} \times 2^{2x-3} = 2^{3x}$ Equating coefficients $2x+4 + 2x-3 = 3x$ $4x+1 = 3x$ $x = -1$	(d) (i) $\sin \theta = -\frac{\sqrt{3}}{2}$ $\theta = -60^\circ \quad -120^\circ$ $\tan \theta + 1 = 0$ $\theta = 135^\circ \quad -45^\circ$

QUESTION 7 If the bearing of the lighthouse, L, from Q is 230°, then the bearing of the point Q from the lighthouse is $(230^\circ - 180^\circ) = 50^\circ$. The boat travels 12 km/h; so in 40 minutes it covers 8 km.	QUESTION 8 (a) $\begin{aligned} \text{LHS} &= (\sec \theta - \cos \theta)^2 \\ &= \sec^2 \theta - 2 \sec \theta \cos \theta + \cos^2 \theta \\ &= \sec^2 \theta - 2 + \cos^2 \theta \\ &= \sec^2 \theta - 1 + \cos^2 \theta - 1 \\ &= \sec^2 \theta - 1 - (1 - \cos^2 \theta) \\ &= \tan^2 \theta - \sin^2 \theta \\ &= \text{RHS} \end{aligned}$
(ii) From $\triangle PQL$, angle $PQL = 180 - (50 + 20) = 110^\circ$.	(b) $\begin{aligned} \text{LHS} &= (x^2 - y^2)^2 + xy \\ &= [(x - y)(x + y)]^2 + xy \\ &= (x - y)^2 + xy & (x + y) = 1 \\ &= x^2 - 2xy + y^2 + xy \\ &= x^2 - xy + y^2 \\ \\ \text{RHS} &= x^3 + y^3 \\ &= (x + y)(x^2 - xy + y^2) \\ &= x^2 - xy + y^2 & (x + y) = 1 \\ &= \text{LHS} \end{aligned}$
(iii) Applying the sine rule in $\triangle PQL$: $\frac{PL}{\sin 110^\circ} = \frac{8}{\sin 50^\circ}$ $PL = \frac{8 \sin 110^\circ}{\sin 50^\circ}$ = 9.8 km to the nearest 100 m (iv) Let R be the point due east of the boat. In $\triangle PLR$: $\cos 20^\circ = \frac{PL}{PR}$ $\therefore PR = \frac{PL}{\cos 20^\circ}$	(c) (i) $u = 3 + \sqrt{7}$ $\therefore u + \frac{2}{u} = 3 + \sqrt{7} + \frac{2}{3 + \sqrt{7}}$ $= \frac{(3 + \sqrt{7})^2 + 2}{3 + \sqrt{7}}$ $= \frac{9 + 6\sqrt{7} + 7 + 2}{3 + \sqrt{7}}$ $= \frac{18 + 6\sqrt{7}}{3 + \sqrt{7}}$ $= \frac{6(3 + \sqrt{7})}{3 + \sqrt{7}}$ $= 6$ (ii) $u + \frac{2}{u} = 6$ $\therefore (u + \frac{2}{u})^2 = 36$

$$= \frac{9.8}{\cos 20^\circ}$$

$$\approx 10.4 \text{ km}$$

$$\therefore \text{Time} = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{10.4}{12}$$

$$= 0.86 \text{ h}$$

$$= 52 \text{ min}$$

$$\therefore \text{Time taken is 52}$$

min and actual time

will be 12.52 p.m.

$$u^2 + \frac{4}{u^2} + 4 = 36$$

$$u^2 + \frac{4}{u^2} = 32$$